iregi-rust benchmark

Batched one-loop tensor reduction in Rust versus IREGI and MadLoop

2026-10-23 · AMD EPYC 9754 · CPUs 0 / 254 · identified kernel snapshots

Reading guide. Every timing ratio is *iregi-rust time ÷ reference time*. Green means faster, white parity and red slower; timings are per phase-space point. Gray cells are unmeasured; amber marks pending or inconclusive evidence. Source inspection does not establish numerical agreement. The latest measured optimized kernel (6b24c03) passes all eight numerical comparisons, with seven timing regressions and one noisy row. It predates the standalone planar API promotion; performance parity remains unmet. Historical sessions and 80949db physical comparisons remain separate.

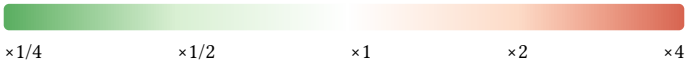
Acceptance gates

Correctness	Agreement with original IREGI and isolated MadLoop graphs	**Pending**
Kernel performance	Equal or better on identical inputs; separate measured snapshots	**Unmet**
Complete evaluation	Six physical rows measured; two failures retained	**Inconclusive**

1 Historical kernel baseline: 4c32ece

Table 1: **Massive bubble/triangle, binary64, hybrid mode.** Rust and IREGI total medians in µs/point. Ratios are paired-block medians; intervals are 99% bootstrap confidence intervals. Batch 1 is latency; batch 32 is amortized cost per point. Each backend uses its own paired reference.

Case	Rank	Interpreter Batch 1	Interpreter Batch 32	Native Batch 1	Native Batch 32
Bubble	0	3.176 µs IREGI 1.988 µs · ×1.598 99% CI 1.585–1.607	1.708 µs IREGI 1.885 µs · ×0.906 99% CI 0.902–0.911	0.936 µs IREGI 0.943 µs · ×0.993 99% CI 0.987–1.003	0.903 µs IREGI 0.878 µs · ×1.031 99% CI 1.024–1.042
Bubble	1	9.552 µs IREGI 5.707 µs · ×1.670 99% CI 1.655–1.688	5.831 µs IREGI 5.748 µs · ×1.015 99% CI 1.008–1.019	4.663 µs IREGI 3.044 µs · ×1.534 99% CI 1.526–1.549	3.231 µs IREGI 3.029 µs · ×1.069 99% CI 1.062–1.081
Triangle	0	29.152 µs IREGI 8.496 µs · ×3.446 99% CI 3.415–3.454	18.229 µs IREGI 8.433 µs · ×2.160 99% CI 2.147–2.172	4.227 µs IREGI 4.968 µs · ×0.849 99% CI 0.844–0.858	8.405 µs IREGI 4.899 µs · ×1.715 99% CI 1.704–1.723
Triangle	1	38.505 µs IREGI 18.297 µs · ×2.101 99% CI 2.088–2.108	25.634 µs IREGI 18.224 µs · ×1.408 99% CI 1.400–1.413	8.809 µs IREGI 10.418 µs · ×0.850 99% CI 0.844–0.854	12.461 µs IREGI 10.213 µs · ×1.220 99% CI 1.206–1.228



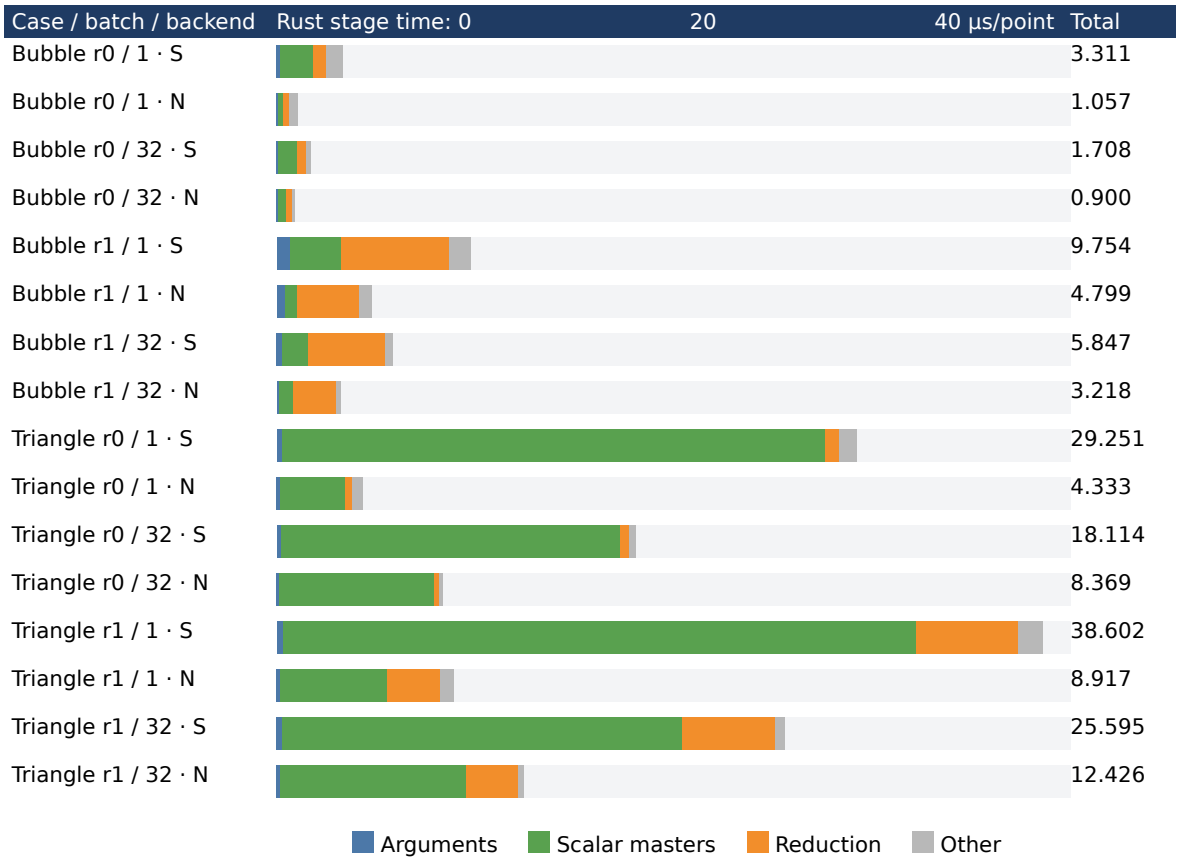
Both backends pass all eight numerical checks at 10^{-10}. The interpreter has seven regressions and one improvement; native has five regressions, one parity result and two improvements. **Absolute IREGI timings changed substantially between sessions**, despite identical inputs/reference binary and CPU settings. Each verdict uses its own paired run; no direct backend speedup is inferred across sessions. Both use the historical `4c32ece` revision; higher-rank performance remains unqualified.

The newer optimized interpreter snapshot `5c9e3dd` passes eight numerical checks, with six timing regressions and two noisy rows. The `80949db` snapshot gives five regressions/three noisy rows. Latest `6b24c03` passes all eight checks: seven regressions/one noisy row. Separate paired comparisons follow; performance acceptance is unmet.

Newer validation includes four integrated graphs. The top pentagon at Float 128/256 agrees with PV/hybrid and MadLoop; original IBP still fails. The latest physical timing comparisons remain inconclusive.

2 Historical stage profiles: 4c32ece

Rust medians from separate instrumented passes, in µs/point. All bars use 0–40 µs/point; S denotes the Symbolica interpreter, N the generated native backend. Sessions are separate: compare each with its own reference on page 1, not with the other backend's absolute bar length. Raw ns/batch are divided by 1000 × batch width.



"Other" is total median minus named stage medians, not a separately timed region. Scalar-master time includes complete master calls; reduction includes stability checks. IREGI has no measured stage split here. For native triangle rank 0, scalar masters cost 3.282 µs at batch 1, but 249.977 µs per batch 32, or 7.812 µs/point: this batch path has higher amortized master cost. The observation identifies a profiling target, not a cross-session speedup or a proven cause. Full stage values and samples remain in the raw data.

2.1 Preparation and measurement conditions

Backend	Bubble r0	Bubble r1	Triangle r0	Triangle r1
Interpreter	1.572 s	0.143 s	0.104 s	0.091 s
Native	0.011 s	0.012 s	0.001 s	0.013 s

Preparation values are single observations without repeated uncertainty, excluded from warm kernel ratios. Batch32 reuses the plan; numerator preparation is outside this tensor-array benchmark.

Both backends of the isolated pushed revision `4c32ece` ran on AMD EPYC 9754 CPU 0, with 5 warm-ups and 21 paired blocks targeting 30 ms, randomized order with seed 271828. Rust used release opt-level3, thin LTO and one codegen unit; Fortran used `-O3`. Structural plans and allocation capacity were reused; numerical results were recomputed, and original recurrence caches were freed per sample. Subprocess transport is excluded. All source/binary provenance checks pass, and every ratio interval is no wider than 2% of its median. Raw samples, inputs, flags and hashes are retained in `validation/benchmark/baseline-4c32ece/` and `validation/benchmark/native-4c32ece/`.

A separate frozen-native policy experiment keeps `Whole`: all 18 C0/D0 comparisons across three orderings find `FixedRows4/8/16` slower, with per-comparison 99% intervals excluding parity. It uses 32 valid lanes, CPU 0 and 21 rotated blocks. This selects a batch policy, not IREGI speedup; see `validation/native-policy/`.

Native allocator adoption remains rejected. Historical 24 timing rows give two singleton parities/22 noisy rows; new D0 timings include two batch32 regressions. D0 exact old/new parity passes 40 configurations/1,488 complex outputs (f64/DD/Float256); interpreter differences remain. Separate candidate decisions appear in the optimization ledger and `validation/native-stack-batch/`.

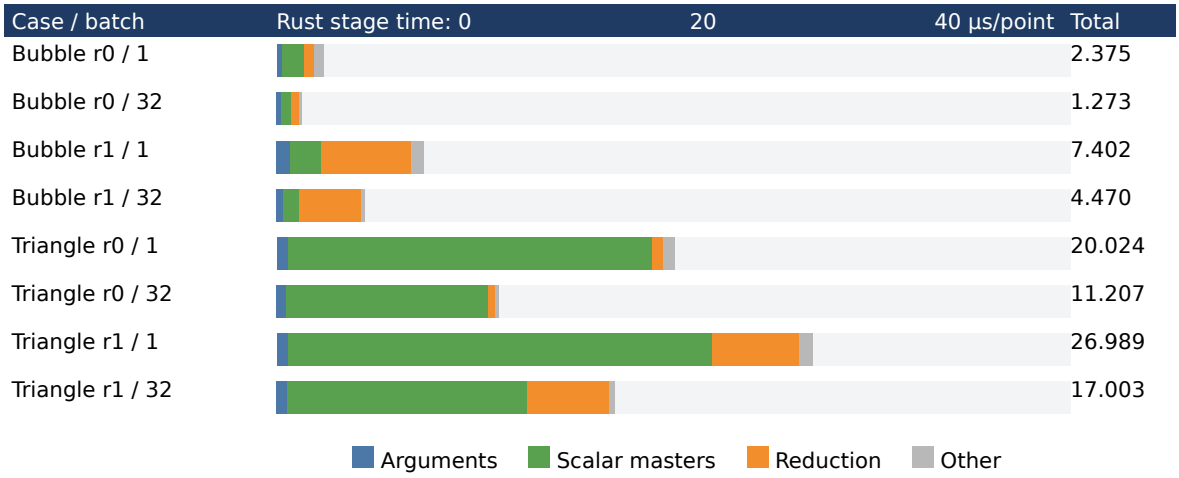
3 Latest kernel snapshot: `6b24c03`

All eight massive bubble/triangle checks pass 10^{-10}, with maximum normalized error 2.46×10^{-13}. Seven timing rows regress; the rank-one triangle singleton exceeds the unchanged 2% interval-width limit (2.357%) and remains **inconclusive**. This measured snapshot predates the promoted standalone planar API and new expression preparation. The performance requirement remains unmet.

Table 2: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Each ratio uses its same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.225 µs IREGI 0.948 µs · ×2.344 99% CI 2.335–2.353 Regression	1.265 µs IREGI 0.896 µs · ×1.413 99% CI 1.407–1.416 Regression
Bubble	1	7.245 µs IREGI 3.048 µs · ×2.388 99% CI 2.372–2.403 Regression	4.498 µs IREGI 3.046 µs · ×1.480 99% CI 1.467–1.488 Regression
Triangle	0	19.618 µs IREGI 4.925 µs · ×3.986 99% CI 3.977–3.996 Regression	11.152 µs IREGI 4.963 µs · ×2.246 99% CI 2.237–2.255 Regression
Triangle	1	26.777 µs IREGI 10.356 µs · ×2.586 99% CI 2.570–2.631 Inconclusive: noise	16.967 µs IREGI 10.411 µs · ×1.627 99% CI 1.621–1.636 Regression

3.1 Separately instrumented component timings



Stages use a separate instrumented pass. “Other” is its total median minus named medians; stage sums are not the uninstrumented gate. Graph/numerator preparation is outside this tensor-array benchmark. The single plan-preparation observations remain in the data, not in speed ratios.

CPU 254, five warm-ups and 41 randomized paired blocks targeting 100 ms use identical inputs, complete tensor arrays and all native Laurent slots. Structural allocations are reused; numerical results are recomputed and reference caches cleared per sample. Transport is excluded. Rust 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`.

The complete optimized workspace build exits successfully. All 414 immutable build-source hashes, both binaries, feature selection and harness match the retained provenance. Other project computations were held. CPU 254 was selected by a read-only load survey because unrelated work occupied CPU 0; external host activity remained uncontrolled. No system settings were changed. Historical sessions used different CPUs and timing protocols; only each session’s paired reference supports its verdict, with no causal revision speedup. Raw samples and reproduction: `validation/benchmark/baseline-6b24c03-licensed/`.

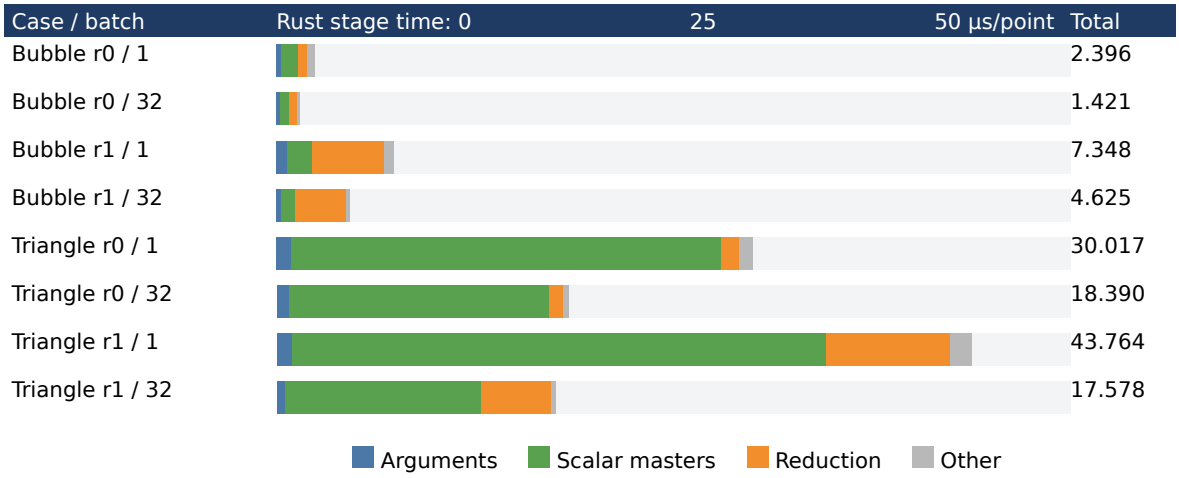
4 Historical kernel snapshot: `80949db`

All eight massive bubble/triangle checks pass the unchanged 10^{-10} criterion, maximum normalized error 2.46×10^{-13}. Five timing rows regress; three exceed the unchanged 2% interval-width limit and remain **inconclusive**. This optimized snapshot includes the adopted master stack allocation, generalized PV and lazy Cayley solve. Later core-allocation and experimental changes are excluded. The performance requirement remains unmet.

Table 3: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Each ratio uses its same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.278 µs IREGI 0.946 µs · ×2.403 99% CI 2.389–2.420 Regression	1.380 µs IREGI 0.892 µs · ×1.550 99% CI 1.537–1.557 Regression
Bubble	1	7.243 µs IREGI 3.083 µs · ×2.342 99% CI 2.318–2.361 Regression	4.619 µs IREGI 3.077 µs · ×1.500 99% CI 1.490–1.518 Regression
Triangle	0	19.553 µs IREGI 4.967 µs · ×3.927 99% CI 3.911–3.944 Regression	17.604 µs IREGI 7.296 µs · ×2.374 99% CI 2.325–2.432 Inconclusive: noise
Triangle	1	43.420 µs IREGI 17.458 µs · ×2.500 99% CI 2.463–2.578 Inconclusive: noise	28.540 µs IREGI 17.009 µs · ×1.672 99% CI 1.490–1.762 Inconclusive: noise

4.1 Separately instrumented component timings



Stages use a separate instrumented pass. “Other” is its total median minus named medians; stage sums are not the uninstrumented gate. Graph/numerator preparation is outside this tensor-array benchmark. The single plan-preparation observations remain in the data, not in speed ratios.

CPU 0, five warm-ups and 21 randomized paired blocks targeting 30 ms use identical inputs, complete tensor arrays and all native Laurent slots. Structural allocations are reused; numerical results are recomputed and reference caches cleared per sample. Transport is excluded. Rust 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-03`.

All 401 immutable build-source hashes, both binaries and the harness match the retained provenance. Other project computations were held; a pending native compiler was suspended. Host activity outside the project was not controlled. The accepted license contains no tracked key material. This is a separate session from the historical tables; its absolute timings establish no revision speedup. Raw samples and reproduction commands: `validation/benchmark/baseline-80949db-licensed/`.

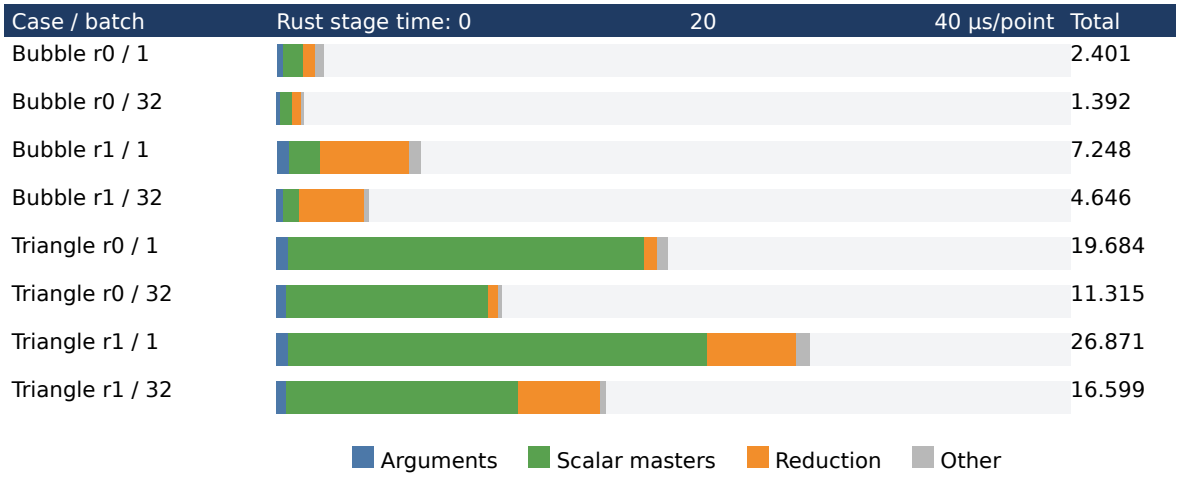
5 Historical kernel snapshot: `5c9e3dd`

All eight massive bubble/triangle checks pass the unchanged 10^{-10} criterion. Six timing rows regress; two exceed the allowed 2% relative confidence-interval width and remain **inconclusive**. Amber cells denote noise, not improvement. This commit predates default stack reuse, generalized PV and the vector-solve candidate. The overall performance requirement is unmet.

Table 4: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Each ratio uses its same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.271 µs IREGI 0.955 µs · ×2.381 99% CI 2.364–2.393 Regression	1.389 µs IREGI 0.901 µs · ×1.541 99% CI 1.535–1.546 Regression
Bubble	1	7.146 µs IREGI 3.019 µs · ×2.367 99% CI 2.362–2.384 Regression	4.593 µs IREGI 3.020 µs · ×1.521 99% CI 1.516–1.532 Regression
Triangle	0	19.619 µs IREGI 4.975 µs · ×3.948 99% CI 3.932–4.027 Inconclusive: noise	11.310 µs IREGI 4.948 µs · ×2.281 99% CI 2.263–2.296 Regression
Triangle	1	26.724 µs IREGI 10.231 µs · ×2.610 99% CI 2.597–2.624 Regression	16.733 µs IREGI 10.279 µs · ×1.620 99% CI 1.608–1.649 Inconclusive: noise

5.1 Separately instrumented component timings



The stage sum belongs to a separate instrumented pass; “Other” is its total median minus named medians. These bars locate costs, not differences between revisions. Preparation is a single observation: bubble r0/r1 0.822/0.112 s; triangle r0/r1 0.067/0.061 s. It is excluded from the warm kernel ratios.

CPU 0, five warm-ups and 21 randomized paired blocks targeting 30 ms use identical inputs and all requested tensor slots. Structural allocations are reused; numerical results are recomputed. Transport is excluded. Rust 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`. All 391 source hashes, both binaries and the harness remain unchanged under the retained provenance checks. Absolute timings from earlier sessions cannot establish a revision speedup. Raw blocks, inputs, manifests and reproduction commands: `validation/benchmark/baseline-5c9e3dd/`.

6 Physical pipeline measurements: `80949db`

Six preserved-input binary64 comparisons pass complete canonical tensors against all three original modes and all three isolated graph Laurent slots against MadLoop. **Every measured timing row remains inconclusive**: its 99% interval exceeds the unchanged 2% relative-width threshold. Both top-pentagon rows fail during preparation; no output or timing substitutes for them.

Table 5: **Equivalent complete numerical pipelines, ms/point.** Repeated captured point; amortized batch32 cost does not establish throughput over varying phase-space samples.

Graph	Ranks Rust/ML	Batch 1	Batch 32
Top triangle	2 / 3	0.877 ms MadLoop 0.108 ms · ×8.114 99% CI 7.808–8.575 Inconclusive: noise	0.729 ms MadLoop 0.105 ms · ×6.918 99% CI 6.817–7.061 Inconclusive: noise
Top box	4 / 4	11.689 ms MadLoop 0.971 ms · ×12.036 99% CI 11.854–12.303 Inconclusive: noise	5.088 ms MadLoop 0.978 ms · ×5.203 99% CI 5.087–5.421 Inconclusive: noise
Massless up box	4 / 4	7.242 ms MadLoop 0.904 ms · ×8.008 99% CI 7.642–8.225 Inconclusive: noise	3.620 ms MadLoop 0.888 ms · ×4.075 99% CI 3.997–4.116 Inconclusive: noise
Top pentagon	4 / 5	Failed before timing Preserved-input conditioning guard	Failed before timing Preserved-input conditioning guard

6.1 Scope and uncertainty

The timers include numerical numerator coefficients, reduction, conversion, contraction, status checks and returned-buffer/cache cleanup. Graph/model and external-wavefunction preparation, input packing, final comparison adapters and process transport are excluded from both. Bare poles and R1 are retained; R2 and UV counterterm graphs are excluded. Routing and rank differences are declared algorithmic choices: **no tensor-stage timing ratio is valid**.

Both engines use Hybrid, 53 working/requested bits, input preservation, CPU 0, five warm-ups and 21 blocks targeting 30 ms. Separate engine runs use an independent bootstrap ratio of medians (10,000 resamples). These intervals describe within-run variation and do not bound drift between engines. Project computations and producer writes were frozen; external host activity was not controlled. Stage samples and raw full responses remain available.

Attempt002 preserves all source/build identities. Attempt001 is retained as unqualified because producer inputs changed during execution. The optimized top-box numerical pass is specific to this captured harness path; earlier CLI binary64 guard rejections remain part of the ledger. Top-pentagon failure is the actual preserved-input derived-distance conditioning diagnostic. No failure is removed from the family count.

Sources, contracts, stage timings and reproduction: `validation/benchmark/physical-joint-quiet/`. The preceding three-block ggH smoke and 80 Python boundary/provenance tests are separately retained in `validation/benchmark/physical-joint-smoke/`.

6.2 Precision follow-up on the failed pentagon point

Promoting the same 95 represented binary64 input components exactly to Float256/512, with requested accuracy53 unchanged, gives convergence over all 280 complex tensor slots within 2.17×10^{-55}. Original PV/hybrid agree within 4.62×10^{-14} and the preserved MadLoop graph within 9.29×10^{-14}. Original IBP still differs by 0.60013; binary64/128 guard failures remain. The first 256-bit driver captured valid outputs before a rational-input comparison-parser failure; the retained separate comparison passes. This is one-point manual precision qualification, not automatic rescue or a timing gate. See `validation/physical/gghhh-guard/`.

7 Current implementation checkpoint

These experiments have separate source and binary identities. They extend the correctness evidence; they do not replace the historical timing tables.

7.1 Ordinary scalar-master storage

The default interpreter now reuses root temporaries through Symbolica's existing allocator. All five families preserve every output, precision and signed zero in 40 configurations each across f64, DoubleFloat and Float256. Shared helpers and Dual preparation are unchanged.

Root temporary slots	A0	B0	dB0	C0	D0
Before	38	503	1,951	220	25,010
After	6	13	24	8	25

The default-stack workspace snapshot compiled with unchanged source hashes before the generalized-PV promotion below. Its current-guard acceptance passes 30 execution, 29 stability, six public mixed-precision and three scalar integration tests.

The optimized `80949db` snapshot also passes eight complete bubble/ triangle kernel checks at batches 1/32, maximum normalized error 2.46×10^{-13}. Three short blocks, zero warm-up and concurrent builds make this a correctness smoke only; its recorded timing ratios are unqualified.

The paired DD E6 experiment preserves all three statuses and 12 complex Laurent outputs exactly, with the independent 10^{-23} check passing in both variants. Each uses order 128 and 3,720 D0 rows.

Diagnostic observation	Before	After
Whole validation duration	46.682 s	43.822 s
Per-child peak resident memory	2,052,724 KiB	558,752 KiB

These are single-run diagnostics, with other compilation active; no timing gate or speed ratio is inferred. The paired experiment uses an earlier, stricter Float guard; fixed-precision DD is unaffected. Current-guard tests are separate. Sources: `validation/stack-reuse/production-factory/`.

7.2 Generalized PV and unresolved precision

Generalized PV for $N \leq 4$ passes nine focused tests and 36 original-IREGI comparisons across 312 complex slots; maximum normalized error is 7.04×10^{-12} at the unchanged 10^{-10} gate. The promoted PV and lazy-solve core also passes five supplemental behaviors: supplied products, multinomial weights, complex scalar/tensor comparisons and Hybrid isolation. The latter uses working128/256, requested53; its binary64 fixture failure is retained. These checks use frozen former-default master archives; combined default-stack integration is separate. Higher points retain IBP/E6. Sources: `validation/generalized-pv/` qualification, promotion and supplemental manifests.

The exact small-routing E6 oracle exposes a remaining production failure:

Working bits	Requested bits	Public result
128	128	Unstable: nonfinite box output
256	128	Unstable: nonconvergent quadrature

Both runs finish within their bounds and expose no successful values. No oracle comparison or requested-128-bit agreement is claimed. The frozen binary predates default stack reuse. See `validation/shifted-pentagon/production-series-qualification.json`.

8 Validation coverage

Table 6: **Evidence ledger.** “Source audited” records code inspection; “verified” identifies executed checks described in that row.

Capability	Required evidence	Status
Independent reference oracle	Untouched/repaired Fortran twins 24 ordinary cases identical; 80 table entries exact; 7 generalized fixtures pass. New pentagon audits reveal reference defects.	Verified
Scalar-master component	OneLoopMaster A0/B0/C0/D0/DB0 Final27 scalar checks and13 consolidated D6 checks pass; all5 mixed physical D0 cases pass atFloat384. Unsupported derivative domains remain explicit.	Verified
IBP component	Powers, dimension shifts and singular identities 21 exact tests; the independent Gamma oracle covers 200 tensor requests and 40 generalized requests in all four slots.	Verified
PV component	Every tensor moment and native Laurent slot Canonical N1–4 ranks 0–7 pass. Generalized N≤4: nine focused + 36 original comparisons pass. Saved N5 rank-seven baseline: 1320 slots. Public N6/N7 tensor ranks pending.	Partial
Precision and Dual	binary64, DoubleFloat, Float, static/dynamic jets Selected f64/DD/Float128–512 and static/dynamic jets pass. Mixed-precision guards: 29 + 6 pass. Small-routing E6 requested at 128 bits: Unstable.	Partial
Execution / CLI / core oracle	Batches through reduction and scalar masters 30 execution and5 argument-guard tests pass. CLI19 smoke checks pass; frozen31 fixtures include30 binary64 plus1 explicit verified rescue.	Partial
Exceptional kinematics	Explicit errors and verified fallback N5 rank-six rescue passes; saved rank-seven baseline passes 1320 slots. Legacy: 22 primary + 8 rescues. E6 failures and broader domains remain visible.	Partial
Graph/numerator preparation	FeynKit DOT to Symbolica coefficients 4 model-boundary, 13 preparation, 4 physical numerator and 19 CLI smoke checks pass.	Verified
Physical MadLoop comparison	Graph-isolated top loops and massless box Integrated ggH, ggHH, ggHHH and massless up-box pass their retained MadLoop comparisons. ggHHH Float128/256 also agrees with original PV/hybrid; original IBP still disagrees and all-mode reports remain failed.	Partial

Snapshot boundary. Counts describe the executed component, public-execution and selected physical checks below. They do not certify all kinematics, higher-point ranks or every physical graph. Open discrepancies remain listed with the original tolerance. Four physical triangle and five massive/complex box cases pass six-component mixed derivative checks. Historical timeouts remain visible. The N5 rank-six public rescue verifies all 210 moments / 840 slots at 128/256 bits, including an independent coefficient. Binary64 Rust and original IBP still exceed the fixed 10^{-10} criterion for that point. The frozen oracle passes 31 cases: 30 at binary64 and one generalized triangle through verified rescue. Broader domains remain pending.

8.1 Known original-reference defect

For shifts $(0,0,0,0)$ and $(0,1,0,0)$, squared masses $(4,9)$, $\mu = 3$ and numerator q_x^4, original PV and hybrid report stability but disagree with an independent Feynman-parameter integral. Their UV residues agree.

Method	Finite coefficient / i
Parameter integral	45.4316300015058951832997…
Original IBP	45.43163000150602
Original PV / hybrid	46.11123944517427

Evidence: `validation/pv-high-rank/evidence.json`. The original PV branch is retained unchanged; the relevant Rust comparisons use original IBP and the independent parameter-integral result.

9 Measurement protocol

Pin release builds to one CPU core and record thread counts, hardware, OS, toolchain, flags, dependency/input hashes, seeds and warm-up. Identify cold or warm caches. Retain raw repeated-block observations, median, dispersion and an output checksum.

Preparation includes graph import, numerator construction and evaluator compilation. Time numerator, reduction, masters and total independently; summed instrumented stages are only diagnostic. Separate library costs from subprocess/file I/O and match physical routing, modes and stage boundaries.

Use identical samples at batch widths 1, 4, 8, 16, 32 and 64, matching numerical precision and including rescue costs. Retain failed lanes. Every speed gate requires correctness: stable binary64 uses the fixed 10^{-10} mixed tolerance on dimensionally normalized outputs. Gate singleton latency and batch-32 throughput per topology/rank/mass family, with IBP, PV and hybrid separate. The target envelope includes seven propagators, rank seven, generalized powers and dimension shifts.

The kernel harness now records every requested configuration before starting subprocesses, preserving interrupted and unrun rows. The latest Python suite passes 80 ledger, provenance, protocol and comparison checks. The optimized joint ggH smoke passes at batches 1/3; its short zero-warm-up timings are diagnostic. The subsequent quiet physical results are reported separately.

Private RealBall certificates pass four paired/eight inherited checks for selected 128-bit real ggH polynomials/jets; other lanes retain ComplexBall. No production or timing gate is claimed.

With the accepted license, 360 batched calls across two workers match sequential A0–D0 values, status and metadata for f64/Float128 and second-order Dual branches. A separate four-worker cold-specialization test passes shared ownership and cleanup checks. These qualify retained SymbolicaBatch snapshots; native kernels remain separate. The earlier rank-one box rejection is retained in `validation/concurrent-workspaces/`.

10 Rebuilding this report

Update `docs/data/results.json` only from retained measurements and rebuild with `bash scripts/build-benchmark-report.sh`. The report uses Libertinus Serif and DejaVu Sans/Mono. Supply their installation directories through Typst's `--font-path` option when needed. Reference checkout revisions are retained in the JSON data. They record the audited source trees and must not be mistaken for a verified dependency build.

10.1 Validation reproduction and remaining checks

Run recorded commands with the pinned environment from `scripts/env.sh`. Exact commands and source hashes accompany each JSON result. The frozen 31-case oracle includes one explicit 128/256-bit rescue; all 30 legacy fixtures pass through 22 primary results and eight verified rescues. Three active rescue, four bubble-Taylor, three scalar API and nineteen CLI smoke checks pass. The new 68-test default-stack acceptance is a separate source scope; earlier failures remain retained.

The final frozen master source passes 27 checks, including extreme-scale B0, six-component scale-only jets in all families, mixed input precision and the full physical D0 derivative batch; see `validation/scalar-masters/qualification.json`. The separate exact factory-parity/default-stack qualification is described above. Generated native kernels remain opt-in; no new timing gate passes.

An independent Symbolica Gamma expansion checks 200 tensor and 40 generalized requests with analytic bubbles disabled/enabled: 480 evaluations and 1920 slots. Nine extracted FeynKit partial-fraction and two planar formula tests pass; public integration remains pending.

N6 inline native-master 128/256-bit arrays converge over 1320 complex slots within 6.46×10^{-29}; public N6/N7 tensor arrays remain unqualified. Public rank-seven probes time out during preparation at 180/300/300 s. The private alias-use candidate passes 73 semantic tests but reaches the 600 s bound in primal preparation (601.010 s with teardown), producing no arrays. Its exact-only proof checks 17 cases, three semantic tests and 25,700 reachable predicates; the separate n-way proof checks 20 cases. Plans have 330 moments, 31,670 aliases and 56 masters. No performance gate follows; evidence and earlier failures remain in `validation/public-highpoint/`.

11 Physical fixture qualification

Single graph	Pointwise numerator	Integrated Rust result
Top triangle: $gg \to H$	Verified	Verified
Top box: $gg \to HH$	Verified	Verified
Top pentagon: $gg \to HHH$	Verified	PV/hybrid + MadLoop
Massless up-quark box	Verified	Verified

11.1 Physical reference contract

Fixtures preserve parameters, actual HELAS polarizations, graph factors and source hashes. The triangle uses two orientations with grouping disabled. MadLoop shifts are $(0, p_1, p_1 + p_2)$ with polynomial variable $x = -Q$; its adapter includes the odd-rank sign. For the FeynKit routing, $N_{\mathrm{FK}(q)} = -\frac{1}{2} P_{\mathrm{ML},1}(p_1 - q)$ with matched couplings/polarizations. Do not apply color or direction factors twice.

Five captures contain 40 graph records, including a second triangle helicity; these are not 40 integrated validations. Thirteen preparation and four selected physical numerator comparisons pass. Actual evaluated momenta are exported because MadLoop may correct the requested point.

Fresh preserved-input Fortran replay matches all 301 numerator coefficients, 903 tensor Laurent entries and 12 contracted slots across four graphs exactly; eight alignment boundary tests pass. Legacy-default ggH is unchanged. This is reference-only qualification; later Rust checks belong to the separate joint run. Equivalent full-graph timing retains Rust/MadLoop ranks 2/3 for ggH, 4/4 for both boxes and 4/5 for ggHHH, and their declared routing. It does not qualify tensor-stage ratios. Sources: `validation/benchmark/physical-alignment/`.

The recorded binary64 $gg \to H$ CLI passes all 15 moments / 60 slots against original IREGI within 1.86×10^{-15} normalized error and MadLoop within 9.38×10^{-15}, including poles. Values and binary identity remain in `validation/physical/ggh-cli-comparison.json` this is not a fresh replay. Nineteen CLI smoke checks pass. Binary64 comparison arithmetic certifies no extra digits from higher working precision; short unpinned timings are excluded.

The historical top-box binary64 CLI rejects: PV fails its cancellation guard and IBP fallback is nonfinite. Float128/256 reruns pass all 280 slots against all three reference modes within 1.15×10^{-13}, and MadLoop within 4.77×10^{-15}. Each freshly prepares/evaluates the numerator. Their normalized tensor difference is 6.09×10^{-37} under binary64 comparison arithmetic. See `validation/physical/gghh-cli-precision-convergence.json`. The Symbolica-backed model adapter passes four boundary, thirteen preparation and four numerator tests while preserving other numerical tokens.

The 80949db massless up-quark box passes all 70 moments / 280 slots against each original mode: maximum error 2.85×10^{-14} against IBP and 7.11×10^{-15} against PV/hybrid. MadLoop finite/single/double coefficients agree within 6.74×10^{-15} with fixed-color factor $-\frac{1}{8}$. Its one-run wall time is unqualified; see `validation/physical/gggg-up-cli-comparison.json`.

Complete top-pentagon CLI reruns at Float 128/256 now agree with original PV/hybrid for all 280 slots within 4.63×10^{-14}, and with MadLoop within 9.30×10^{-14}. Between-run tensor difference is 7.79×10^{-17} under binary64 comparison arithmetic. Original IBP still differs by 0.60014, so the all-mode report remains failed. See `validation/physical/gghhh-cli-precision-convergence.json` and linked reports. Earlier PV/hybrid replay of 378 captured MadLoop slots was exact.

11.2 Physical derivative qualification

Four triangle cases at 384 bits cover real subthreshold/absorptive and complex-mass kinematics, each with six first, second and mixed Dual components. Finite differences converge with maximum final scaled error 5.47×10^{-28}. One equal-complex D0 case also passes, with error 1.11×10^{-34}. The complete five-case D0 batch now passes at 384 bits, maximum final scaled error 2.53×10^{-30} under the unchanged 2^{-70} criterion. Its 224.83 s validation duration is not a benchmark; earlier 180 s timeouts remain retained. See `validation/scalar-masters/physical_box_full.json`. Exact commands and scope are retained in `crates/iregi-masters/tests/validation_evidence.json`.

12 Open qualification ledger

Unresolved item	Observed evidence and remaining check
Binary64 baseline	Historical Rust/IBP errors 2.23×10^{-10} / 3.40×10^{-10} fail. Rank-six rescue passes at the retained point.
Pentagon coverage	Rank-six rescue passes 840 slots; saved rank-seven baseline passes 1320. Later precision/routing edits and broader domains require qualification.
Physical pentagon	Float 128/256 agrees with PV/hybrid and MadLoop; original IBP disagrees. Binary64 and all-mode acceptance remain incomplete.
Shifted scalar	Thirteen standalone checks and public f64 fixtures pass. Small-routing E6 remains Unstable at requested 128; higher/mixed Dual and broader domains remain unqualified.
Derivative domains	Five-case mixed D0 checks pass 2^{-70}. Thresholds and unsupported transverse massless/ Gram/Cayley domains retain errors.
Physical top box	Historical binary64 guard rejection remains visible; Float 128/256 and preserved 80949db binary64 checks pass at retained points.
Supplied-product N6	Certified scalar working 128/256, requested 53 passes with zero poles and invalid-lane isolation; binary64 rejects. Independent difference 1.53×10^{-23} is below estimated (nonrigorous) 6.11×10^{-22}. N7/tensors/mixed Dual remain pending.
Legacy and precision	All 30 retained cases pass: 22 primary and 8 rescues. Three false-success cases are repaired; conservative active-master error estimates remain noncertified. Broader domains remain unqualified.
Performance	Historical 4c32 interpreter: 7 regressions/1 gain; native: 5 regressions/1 parity/2 gains. 5c9: 6 regressions/2 noisy; 80949: 5/3. Latest 6b24: 7/1. Physical: six noisy/two failures. Overall acceptance is unmet.

12.1 Optimization and experimental evidence

Candidate	Executed evidence and decision
Adopted allocation / guards	Core allocation: 67 current opt2 checks; quiet paired run preserves 92 outputs. Guard context: 76 checks and 120 snapshots, 14 cleanup checks; six isolated gains/six noisy rows. Combined 6b24 optimized IREGI gate remains seven regressions/one noisy row.
Roundoff cache	29/37 stability checks and 120 snapshots pass. Candidate rejected: two regressions/ten noisy rows. Separate accuracy-guard diagnostic: two regressions/six noisy rows.
CLI coefficients	72 ordinary and 12 mixed-precision batch comparisons pass across four graphs. Eight opt2 timing families: five gains/one parity/ two noisy. ggHH batch 32 ratio 1.0367, 99% CI 1.0237–1.0501 retains possible regression; unadopted. All 168 paired outputs match.
Native candidates	Dense f64 B0/C0: two gains/six singleton parities/four batch32 regressions. D0 allocator: three singleton gains/two batch32 regressions/one noisy. Both rejected. D0 singleton source is unchanged; its timing gains do not establish allocator benefit.
Certificate candidates	Slim RealBall module: 23 checks, including 8 baseline controls. Separate exact-distance receipts: 49 checks and 120 exact snapshots against the same private RealBall baseline; argument rows unchanged. Neither adopted nor timed.
N5 planar API	Promoted for unit powers, with no automatic fallback. Rank 7, requested 53 bits: ordinary Float128/256 passes 1320 slots and convergence. Mixed Float128 passes all 5280 four-component slots per accepted duplicate lane against Float256 boxes. Transverse jets reject; 9 focused and 49 ordinary tests pass; earlier 600 s timeout remains. Twenty coefficient cases with original Fortran scalar leaves pass; those do not add public native-master coverage. No new performance gate.
Near-vacuum adapter	Private v3 on the 6b24 release dependency closure: six runs pass 96 component matches, dispatch contracts and the five-box contour probe. Targets: f64 at 40 bits, DD at 80, Float128 at 100, Float256 at 128. Production D0/E6 is unchanged.

New CLI/native timings use CPU 254 and 21 paired 30 ms blocks, with project compute held; these isolated candidates are not IREGI or end-to-end comparisons. Historical native 24 rows remain separate. Full paths, scopes and failures are in `docs/data/results.json` effects are not multiplied.

13 Independent pentagon adjudication

These checks establish selected mathematical values independently of an original-reference stability flag. They do not qualify every pentagon tensor, physical graph, runtime domain or numerical precision.

13.1 Rank-six Euclidean tensor

For squared masses $(4, 9, 16, 25, 36)$, $\mu = 3$ and numerator $q_0^5 q_3$, direct simplex integration gives native finite coefficient $-0.0024283789354471 23i$. Independent coordinate mappings agree within 4.34×10^{-19}; reported quadrature error is at most 2.77×10^{-15}. The bound $F \geq 3.75$, angular integration and power counting exclude poles. This uses no reduction/master library; numerical error estimates are not interval certificates. See `validation/pv-high-rank/pentagon-simplex-oracle.json`. Historical binary64 errors remain 2.23×10^{-10} for Rust and 3.40×10^{-10} for IREGI. Public rescue flags the primary unstable and verifies all 840 slots at 128/256 bits plus this coefficient below 10^{-13}. `validation/rescue-pentagon-rank6.json` retains the binary and assertions; full arrays were not printed and its exact build-source snapshot was not saved.

The coherent vendored Symbolica baseline passes four internal scope tests, 57 focused core tests and all 1320 rank-seven slots at 128/256 bits, including the independent coefficient. Its saved source predates later precision/routing edits. Current workspace compilation passes with all 329 source/manifest hashes unchanged; later runtime checks remain separate. The 109.44 s cold run is diagnostic, not a performance benchmark. See `validation/symbolica-production-qualification.json`. The earlier 240 s timeout and 115.35 s isolated-patch run remain historical.

13.2 Physical canonical moment 54

For $gg \to HHH$, rank-four moment $q_0 q_3^3$ has UV degree -2; its UV pole must vanish. Original IBP instead returns approximately $3.33691766067 \times 10^{-6} i$ in native normalization. The 128/256-bit PV evaluation suppresses its OneLoop-normalized UV residual from 7.9×10^{-31} to 2.3×10^{-69}, with both IR slots zero. Its converged OneLoop finite coefficient is approximately $3.3474440772594743330 \times 10^{-13} + 2.9254797223941252086 \times 10^{-13} i$. This supports original PV/hybrid and rejects the original IBP result for this component. It is not a full physical tensor comparison. Raw precision evidence is in `validation/pv-high-rank/canonical-pentagon-precision.json canonical-pentagon-adjudication.json` records fixed-gate comparisons.

13.3 Standalone six-dimensional scalar

Use $J_1 = \frac{I_6}{i\pi^3}$, shifts $(0, e_0, e_1, e_2, e_3)$ and squared masses $(10, 12, 15, 18, 22)$. Here $F \geq \frac{39}{4}$, $\det R = 2600$ and $\det G = -16$.

Method	Finite J_1
Independent simplex	$-0.000177140138264008180607652131 0$
Original scalar/general APIs	$-0.000149530250241284 55$; every mode reports stable

Increasing quadrature order and precision through 70 digits converges to the first value. Independent integration of ordinary D4 values over a common mass shift, with a separately integrated tail, agrees within 4.93×10^{-15}. A box-only half-line implementation also agrees to approximately 8.42×10^{-17} or better in the recorded checks. Ordinary D4 and coalesced D4/D6 normalization checks pass. Evidence and reproduction source are in `validation/shifted-pentagon/evidence.json`. Thirteen consolidated D6 tests cover eight genuine-master checks and five endpoints. Retained public f64 fixtures pass finite values, poles, scaling, rejected endpoints and lane isolation; the earlier quadrature-budget failure remains historical. Pre-vendor public DoubleFloat and Float128 tests now pass three scaled lanes at requested 80-bit accuracy and 10^{-23} normalized oracle tolerance. First common-mass Dual primal/derivative checks pass at 2×10^{-14}, with zero poles. The DD/Float validation durations total 165.69 s, exceeding the historical 150 s combined timeout; this is no performance benchmark. Build identities and limits are in `validation/shifted-pentagon/public-core-qualification.json`. A separate vendored paired DD E6 test passes exact stack-allocation parity. Full requested 128-bit accuracy fails for the new small-routing fixture; higher/mixed Dual, broader endpoints and native E6 remain unqualified. The original complex-box reference discrepancy is documented in `validation/shifted-pentagon/contours.json`.

14 Reference conventions

These conventions were read from the supplied original sources. Their presence in this report is not a claim that every adapter has passed its numerical tests.

14.1 Kinematics and scale

Use Minkowski signature $(+,-,-,-)$ with component order (p^0,p^1,p^2,p^3). Propagators are $D_j = \left(q+p_j\right)^2 - m_j^2 + i0$, raised to positive integer powers. Generalized integrals use $d = 4 + \mathrm{idim} - 2\varepsilon$ and the native raw measure; dimension shifts introduce powers of π. Native tensor reduction accepts one shift per propagator and removes the first common shift internally, then translates tensor coefficients back. The MadLoop interface instead receives the nonzero shifts and inserts a zero first shift. A reference comparison must preserve the chosen numerator routing when it changes the propagator direction.

The original `MU` / `MUR` argument is an energy scale, not its square. Its scalar wrapper passes μ^2 to QCDLoop and μ to OneLOop. Complex mass-squared inputs must retain their imaginary part; the reference switches to its OneLOop backend for complex masses. OneLoopMaster accepts real external squared invariants, complex squared masses with $\Im(m^2) \le 0$, and a positive real squared scale. Its continued logarithms and polylogarithms own the Feynman branch prescription; no independent branch implementation belongs in the reduction adapter. Preserve inputs by default; legacy on-shell snapping must be requested explicitly. Dual derivatives follow the selected analytic branch away from nondifferentiable boundaries.

14.2 Native poles and MadLoop conversion

The native result order is

$$(U, I, J, F) = (\varepsilon_{\mathrm{UV}}^{-1}, \varepsilon_{\mathrm{IR}}^{-1}, \varepsilon_{\mathrm{IR}}^{-2}, \mathrm{finite}).$$

These symbols label separate coefficients. In particular, a scaleless integral can have canceling UV and IR contributions even when its combined Laurent series is zero. Three combined coefficients do not determine their separate origin. OneLoopMaster uses $(\mathrm{finite}, \varepsilon^{-1}, \varepsilon^{-2})$ ordering.

The original `IMLOOP` adapter combines the native poles and maps a tensor of homogeneous rank r to MadLoop order (finite, simple pole, double pole). Defining $a = \gamma_E + \ln\pi$, $b = \frac{a^2}{2} + \frac{\pi^2}{12}$ and $c_r = \frac{(-1)^r}{i\pi^2}$, its explicit mapping is

$$(F_{\mathrm{ML}}, S_{\mathrm{ML}}, D_{\mathrm{ML}}) = c_r(F + a(U+I) + bJ, U + I + aJ, J).$$

For a OneLOop-normalized scalar master (L_0, L_1, L_2), the scalar wrappers use

$$J = i\pi^2 L_2, \quad U + I = i\pi^2(L_1 - aL_2),$$

$$F = i\pi^2\left(L_0 - aL_1 + \left(\frac{a^2}{2} - \frac{\pi^2}{12}\right)L_2\right).$$

The UV/IR split requires independent integral classification. In particular, the scaleless bubble is $(U, I, J, F) = (i\pi^2, -i\pi^2, 0, 0)$ in the original wrapper. Applying the MadLoop conversion to an already normalized master result would count normalization twice.

14.3 Polynomial and tensor interfaces

A rank bound R has $\binom{R+4}{4}$ Cartesian coefficients in the original four-component basis. `ti_reduce::calc_pos` traverses total degree in ascending order, then exponents of q^0, q^1, q^2 in descending order; the exponent of q^3 is the remaining degree. Thus the start is $\left(1, q^0, q^1, q^2, q^3, (q^0)^2, q^0q^1, q^0q^2, q^0q^3, (q^1)^2, \ldots\right)$. Numerator preparation and tensor output must share this order. The original native and MadLoop tensor conventions differ in multinomial factors: the native path multiplies symmetric components by the factor returned by `all_integers`, whereas `ML5_CONVENTION=.TRUE.` uses unweighted components. The independent oracle sets this flag **without** applying `IMLOOP`, preserving the native four-pole normalization. Mixed-monomial tests detect multiplicity mismatches.

Reduction modes are 0: IBP, 1: PaVe, 2: PaVe with IBP stability improvement. Direction mode 2 reverses the propagator order and shifts the loop origin. The source labels direction modes 4/5 as unimplemented quadruple precision; their numbers are not evidence of high-precision support.

15 Algebra reuse audit

All symbolic algebra belongs to Symbolica and the FeynKit / Spenso / Idenso ecosystem. Before introducing a missing operation, record three checks: Symbolica APIs and implementation; Symbolica usage/examples/tests; and FeynKit/gammaloop facilities and tests. An adapter or data-layout conversion must not grow an independent polynomial, simplification or matrix engine.

15.1 Existing facilities to reuse

- **Tensor identities and contractions:** Idenso contracts metrics and reduces supported gamma/color identities; Spenso carries tensor structure. Keep Lorentz dimension and any gamma-five prescription explicit.
- **Coefficient extraction:** FeynKit's tensor reducer uses Idenso contraction without global expansion, `to_dots()` and the shared `coefficient_list` collector. Preserve scalar factorization until polynomial extraction requires expansion.
- **Propagator and scalar-product algebra:** FeynKit's `IntegralFamily` uses Symbolica for polynomial conversion, exact rank and linear solves, basis completion and mappings. Its Gaussian-rational conversion avoids statistical expression-field zero tests that may discard small coefficients.
- **Numerical preparation:** Compile reusable Symbolica expressions outside the sample loop. Evaluation and scalar-master types must preserve the selected precision and all Dual components without binary64 round trips.
- **Laurent coefficients:** IBP/PV factors and native normalization use cached Symbolica `series` templates and `coefficient` extraction. The three reuse checks are the pinned Symbolica `src/atom/core.rs` and `src/poly/series.rs` APIs; its `src/derivative.rs` pole-series tests; and existing use in `vakint/src/projection.rs` and OneLoopMaster's `src/sheet_exact.rs`. No separate Laurent convolution engine is needed. Symbolica also evaluates the independent Gamma/series triangle oracle.

15.2 HEPKit tensor-reduction optimizations

The local `feynkit-tensor` reducer stores repeated-vector contractions as symmetry orbits instead of factorially many labeled pairings. Its joint-color dynamic program builds orbit intersections; alternating cycles build orbit Gram matrices; Symbolica solves the smaller orbit system. A rank-20 isotropic projection can have one orbit instead of 19!! labeled pairings.

The implementation caches canonical Weingarten coefficients and has a closed isotropic path. General projectors are validated through rank 20; the special all-equal isotropic route reaches rank 32. These are upstream limits, not a statement of iregi-rust's current supported domain.

For denominators with external momenta, declare an independent external basis and average only the transverse space. Odd total tensor ranks need not vanish. Keep D symbolic while constructing high-rank projectors; early integer specialization can make the universal metric basis singular. External Gram inverses can amplify small-Gram numerical errors, so projector preprocessing requires a conditioning-aware comparison against direct recurrences.

This projector reduces tensor numerators, not complete scalar integral families. Reuse it where applicable and benchmark the resulting plan; no factorial-count argument alone establishes a numerical speedup. Retain explicit output/pairing limits and report their failures honestly.

15.3 References and provenance

The untouched generalized IREGI entry point stops during factorial-table initialization: the configured powers/rank request factorial 16, while the helper rejects arguments above 12. A narrow repair in a generated reference copy replaces that table expression by its exact binomial recurrence. The untouched/repaired twins agree exactly on 24 ordinary cases; all 80 table entries match independent integer binomials; seven generalized fixtures check normalization, unit powers, mass derivatives and a tadpole dimension shift. The original `STOP` returns exit code zero, so the harness also requires an explicit result record. Evidence lives in `validation/reference/fixtures.json` and the report's JSON snapshot.

OneLoopMaster documents corrected analytic-continuation defects in its upstream OneLOop reference, including complex-mass box cases. Preserve each disagreement as a visible case and adjudicate it with independent analytic, high-precision or parameter-integral evidence. Agreement must not mean reproducing a known incorrect branch.

Primary references: Alwall et al., appendix C.1 HEPKit TensorReducer Goode et al., tensor projectors Collins–Matsumoto, Weingarten recurrence NinjaRS report style. Local source audit: FeynKit's `feynkit-tensor/src` projector modules and `feynkit-graph/src/integrals.rs` original IREGI's `IREGI4ML5_interface.f90`, `ti_reduce.f90`, `si_reduce.f90`, `mis_warp.f90` and `funlib.f90`.